INCH-POUND

MIL-B-53107(ME)
23 February 1990

MILITARY SPECIFICATION

BRIDGE, LIGHT ASSAULT: 23 METER SPAN MLC 30

This specification is approved for use within the USA Belvoir Research, Development and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a 23 meter (75 foot), double fold, scissoring type bridge which, together with the launcher (MIL-L-53106), constitute the light assault bridge system.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

QQ-S-781	- Strapping, Steel and Seals.
PPP-T-60	- Tape, Packaging, Waterproof.

MILITARY

MIL-P-116	- Preservation, Methods of.
MIL-T-704	- Treatment and Painting of Material.
MIL-H-6083	- Hydraulic Fluid, Petroleum Base for Preservation and Operation.
MIL-B-43436	- Band, Marker, Blank.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research, Development, and Engineering Center, ATTN: STRBE-TSE, Fort Belvoir, VA 22060-5606 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5420

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MIL-C-46168 - Coating, Aliphatic Polyurethane, Chemical Agent Resistant.
MIL-E-23765/2 - Electrodes and Rods - Welding, Bare, Solid, Low Alloy Steel.
MIL-L-53106 - Launcher, Bridge, Light Assault Towed.

STANDARDS

FED-STD-H28 - Screw Thread Standards for Federal Services.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-130 - Identification Marking of US Military Property.
MIL-STD-209 - Slinging Eyes and Attachments for Lifting and Tying Down Military Equipment.
MIL-STD-454 - Standard General Requirements for Electronic Equipment.
MIL-STD-781 - Reliability Tests, Exponential Distribution.
MIL-STD-810 - Environmental Test Methods and Engineering Guidelines.
MIL-STD-882 - System Safety Program Requirements.
MIL-STD-889 - Dissimilar Metals.
MIL-STD-1472 - Human Engineering Design Criteria for Military Systems, Equipment and Facilities.
MIL-STD-1474 - Noise Limits for Army Materiel.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATIN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

ME

TA 13226E5250 - Bridge, Light Assault, Military Land Class 30.
D13226E5306 - Cylinder Hydraulic Bridge Hinge.

(Copies of drawings required by contractors in connection with specific acquisition functions should be obtained from the USA Belvoir Research, Development, and Engineering Center, ATIN: STRBE-JB, Fort Belvoir, VA 22060-5606.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of

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the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Boiler and Pressure Vessel Code, Section IX, Welding Qualifications. (Application for copies should be addressed to American Society of Mechanical Engineers, 345, East 47th Street, New York, NY 10017.)

AMERICAN WELDING SOCIETY, INC. (AWS)

AWS A5.1 - Electrodes, Arc Welding, Covered Carbon Steel.
AWS A5.5 - Electrodes, Arc Welding, Low Alloy Steel Covered.
AWS A5.10 - Welding Rod and Electrode Aluminum and Aluminum Bare.
AWS A5.18 - Welding, Carbon Steel Filler Metals for Gas Shield Arc.

(Application for copies should be addressed to the American Welding Society, 2501 N.W. 7th Street, Miami, FL 33125.)

(Non-Government standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, (except for associated detail specifications, specification sheets or MS standards), the text of this document shall take precedence. Nothing in this document, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The bridge and bridge components shall be in accordance with top assembly 13226E5250 and as specified herein.

3.1.1 Drawings. The drawings forming a part of this specification are end product drawings. The contractor is responsible for preparing his own shop drawings. Where tolerances prescribed could cumulatively result in improper fits, the contractor shall provide tolerances within those prescribed on the drawings to insure proper fit, assembly, and operation of the bridge. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Any data (e.g., shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by this contractor or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or his designated representative.

3.2 First article. Unless otherwise specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.2.1 Hydraulic system components. In addition to the first article bridge the following hydraulic components shall be furnished and tested by the contractor prior to the time the first articles are furnished: One each of all

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hydraulic fittings, hose assemblies, valves, quick-disconnects, couplings and plugs, hydraulic manifolds, and hydraulic cylinders. The hydraulic components shall be identical to those furnished on the first article. Tests shall be those specified in section 4 and shall be subject to surveillance and approval by the Government.

3.3 Material. Material shall be as specified herein and as shown on the applicable drawings. The surfaces of materials used shall be free from foreign matter, scale, rust, or corrosion.

3.3.1 Material deterioration prevention and control. The bridge shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operating and storage environments to which the bridge may be exposed.

3.3.1.1 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

3.3.1.2 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with component and subcomponent, and shall make information available upon request to the contracting officer or designated representative.

3.3.1.3 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces, and parts incorporated in the bridge may be newly fabricated from recovered materials to the maximum extent practical, provided the bridge produced meets all other requirements of this specification. Used, rebuilt or remanufactured components, pieces, and parts shall not be incorporated in the bridge without prior approval of the contracting officer.

3.3.2 Welding electrodes. Unless otherwise specified on the end product drawings, the following welding electrodes shall be used.

3.3.2.1 For aluminum. Welding electrodes for aluminum shall conform to AWS A5.10, classification ER5356.

3.3.2.2 For steel.

3.3.2.2.1 High-strength steel. Welding electrodes for high-strength steel, including joints with steel castings, shall conform to AWS A5.5, classification E12015, E12016, E12018M, or MIL-E-23765/2, classification MIL-1205-1.

3.3.2.2.2 Mild steel. Welding electrodes for mild steel, including joints with steel castings, shall conform to AWS A5.1, classification E7015, E7018, E7028, OR AWS A5.18, classification E70 S-3.

3.4 Hydraulic oil. Oil for all hydraulic systems shall conform to MIL-H-6083.

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3.5 Bridge hydraulic system. The assembled hydraulic system shall withstand pressures as specified in 3.5.1, 3.5.2, and 3.5.3 without external leakage (except leakage normally expected at piston rod packing), permanent deformation or rupture to any part of the system. Prior to delivery, the hydraulic system shall be filled with hydraulic oil conforming to MIL-H-6083. The hose age shall not exceed 12 quarters from the date of assembly to the date of acceptance of the bridge by the Government. Separately procured hose assemblies shall be provided with suitable plastic protectors of end openings and threads.

3.5.1 Hydraulic cylinders. When tested in accordance with 4.5.2.4, the hydraulic cylinders shall withstand the pressure specified on the end item drawing without external leakage, permanent deformation or rupture of any part of the system.

3.5.2 Fittings and hydraulic hose. When tested in accordance with 4.5.2.5, fittings and hydraulic hose shall conform to the individual end product drawing requirements and shall withstand the pressure for the system of which they are a part.

3.5.3 Quick-disconnect couplings. When tested in accordance with 4.5.2.5, quick-disconnect sockets and plugs shall conform to the individual end product drawing requirements and shall withstand the pressure specified for the system of which they are a part.

3.5.4 Relief valve and counterbalance valves. When tested in accordance with 4.5.2.6, the relief valves and counterbalance valves shall be as specified on the applicable end product drawings. In addition, the valves shall withstand and operate at the applied pressure without malfunction or failure.

3.5.5 Internal leakage. When tested in accordance with 4.5.2.7 the hydraulic oil leakage during hydraulic cylinder pressure test shall not exceed the rate specified in drawing 13226E5306.

3.6 Threads. All threads not otherwise covered by this specification or the end product drawings shall conform to FED-STD-H28.

3.7 Hydraulic system cleanliness. Prior to installation on the bridge the interior and exterior surfaces of all hydraulic components shall be washed and flushed to remove all visible contaminant particles and foreign elements. Openings of the components shall be immediately covered until installation. The hydraulic system shall be operated and bled prior to being subjected to the tests specified herein. The hydraulic system shall be kept clean of foreign matter and contamination during all tests and when tested in accordance with 4.5.2.8 shall meet the contamination limits specified in table I prior to acceptance.

Table I. Contamination limits (particles per milliliter).

Quantity (max)	Particle Size
1000	Greater than 10 microns
100	Greater than 20 microns

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3.8 Performance. When tested in accordance with 4.5.2.9, the bridge shall perform as specified herein without evidence of any of the following defects:

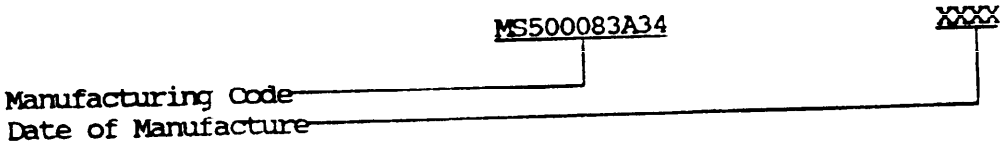
a. Malfunctioning of components or failure of bridge to unfold and fold as required.
b. Binding of parts during launching and retrieving.
c. Permanent distortions (other than normal wear).
d. Material defects such as cracks or open joints.
e. External hydraulic leakage.
f. Failure of hydraulic systems to operate as required.

3.9 Lubrication. Lubrication shall be as shown on the end product drawings or as specified herein.

3.10 Cleaning, treatment, and painting. Cleaning, treatment, and painting shall be as shown on the end product drawings. Where MIL-T-704 paint is required, the finish coat shall be camouflage green 383 conforming to MIL-C-46168.

3.11 Identification marking. The bridge components shall be identified in accordance with MIL-STD-130 and as indicated on the end product drawings.

3.11.1 Identification marking for hose assemblies. Each hose assembly shall have identification marking stamped on an aluminum band conforming to MIL-B-43436, except that the band shall be of sufficient width to allow a minimum of two lines of 1/8-inch metal stamping. Unless otherwise specified on the drawings, the marking band shall contain the following information.



3.12 Stenciling. The nomenclature, national stock number, weight, and cubage shall be stenciled on each bridge. The letters "NET WT" shall precede the numerals in the weight and the letter "CU" shall precede the cubic displacement. Markings shall be gothic-type capitals and Arabic numerals 3/4 inch high. Stenciling shall conform to MIL-T-704.

3.13 Government-loaned property. Unless otherwise specified (see 6.2), the following property in the quantities indicated, will be loaned by the Government (see 6.5):

Item No.	Quantity	Description
1	1 set	Inspection Equipment for Bridge.
2	as required for IPT	Launcher and Tow Vehicle for Bridge.
3	1	Light Assault: 23 Meter Span MLC 30.
4	1	Test Fixture, Hydraulic Cylinder.
5	1 set	Technical Manual.
		Quality Assurance Provisions.

3.14 Reliability. The specified mean-time-between-failure (Oo) shall be 20 cycles when tested in accordance with MIL-STD-787, as specified in 4.5.2.10.

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3.15 Maintenance ratio. The ratio of man-hours of maintenance required to the hours of operation performed shall not exceed 1.0.

3.16 Slinging and tiedown provisions. Slinging and tiedown provisions shall be as shown on the drawings and shall be tested in accordance with 4.5.2.11 to meet the requirements of MIL-STD-209.

3.17 Interchangeability. All parts, components and assemblies having the same identification number shall be functionally and dimensionally interchangeable. Interchangeability is defined as two or more parts (components and assemblies) possessing such functional and physical characteristics as to be equivalent in performance and durability, and capable of being exchanged one for the other without alteration of themselves or of adjoining parts (components and assemblies), except for built-in adjustment, and without selection for fit or performance.

3.18 Safety. The bridge and associated subsystems shall not contain uncontrolled catastrophic or critical hazards as defined in MIL-STD-882. All exposed parts which are subject to high operating temperatures or which are energized electrically shall be insulated, fully enclosed or guarded. Rotating parts that are a hazard to operating personnel shall be guarded. Protective devices shall not impair the operating functions of the equipment. Safety warning plates and operating instruction plates shall be affixed to the equipment in conspicuous locations to inform and provide the adequate degree of protection required. Design requirements of MIL-STD-454, MIL-STD-1474 apply.

3.19 Human engineering. The light assault bridge shall conform to human factors engineering design criteria as described in MIL-STD-1472. Special emphasis shall be given, but shall not be limited to 4 (General Requirements), 5.4 (Controls), 5.5 (Labeling), 5.9 (Design for Maintainability), and 5.13 (Hazards and Safety) of MIL-STD-1472, as applicable.

3.20 Rail impact. The bridge shall be mounted on the launcher and mounted on a railway flatcar as specified in MIL-STD-810. The bridge shall then be tested in accordance with 4.5.2.12.

3.21 Specification-control items. Prior to using any specification-control item other than a suggested source item, the contractor shall furnish to the contracting officer evidence that the item conforms to all requirements on the applicable drawing. The contractor is responsible for performing all inspections and tests on specification or source control items if not performed by the vendor. No change shall be made to any source or specification control item after approval of the first article model without prior approval by the contracting officer.

3.22 Workmanship. All members, components, and assemblies containing moving parts shall be assembled free of harmful extraneous materials such as metallic particles, dirt, grit, or corrosion.

3.22.1 Castings. All casting used on the bridge shall be cleaned of harmful extraneous material, such as sand, dirt, pits, scale, and flux. Rework shall be limited to procedures which do not reduce strength or affect function.

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3.22.2 Metal fabrication. Metal used in fabrication shall be free from kinks and sharp bends. The straightening of material shall be done by methods that will not cause injury to the material. Corners shall be square and true. Flame-cutting, using tips suitable for the thickness of the steel, may be employed instead of shearing and sawing for non-maraging steel only. Neither aluminum alloy, nor maraging steel shall be heated for forming or cut by methods using heat. All bends shall be made with controlled means to insure uniformity of size and shape. Precaution shall be taken to avoid overheating. Heated steel shall be allowed to cool slowly. All parts shall be completely and accurately shaped to the drawing requirements prior to joining (welding). Joints, butts, and splices shall be limited to those shown on the end product drawings. External surfaces shall be free of butts, sharp edges and corners, except when sharp edges or corners are required or where they are not detrimental to safety.

3.22.3 Welding. Unless otherwise specified on the product drawings, welding shall be done in accordance with the requirements of this specification. Prior to the fabrication of any weldment, the contractor shall prepare and submit to the contracting officer for approval, an isometric or perspective drawing showing the location of each joint and the joint welding procedures to cover all welding, including a general outline for the repair of base material and welded joints. These procedures shall also address the proposed method of surface preparation prior to welding. The procedures shall be qualified in accordance with the ASME Boiler and Pressure Vessel Code, Section IX, except that the guided bend test is not required for steel castings. All welds to be qualified shall be made on the same alloy, gage and temper of material, with the same filler materials, and in the same position as will be used in production. Any change in procedure during fabrication shall require requalification in accordance with the requirements of the ASME Boiler and Pressure Vessel Code, Section IX.

3.22.3.1 Aluminum welding. The aluminum surfaces of parts to be welded shall be free from scale, paint, grease, oxide film and other foreign matter. The welding shall be done by the gas-metal-arc (MIG) method, except as shown on end product drawings. Where material gage or size prevents use of MIG welding, the gas-tungsten-arc (TIG) method may be employed. Welds may be ground, filed, buffed, or chipped, but not hammered, and shall be free from defects, such as base metal cracks adjacent to or behind welds, cracks in weld metal, under-cutting of base metal, lack of fusion with parent metal or between parts, porosity, lack of penetration, lack of prescribed fit, inclusions, or spatter. To assure that all welding variables have been addressed, the welding procedures required by 3.22.3 shall include provisions for a daily on-the-job aluminum filler weldbreak test by each productive welder during each work shift. The procedure shall require that the welder construct the "weldbreak" specimen using the same materials, alloys, and cleaned, and prepared in accordance with the applicable procedures and techniques developed for the end item weldment being welded at the time the break test is required. Weld specimens shall be prepared using two 3/8- X 4- X 4-inch plates, welded, using a 3/8-inch fillet weld. The procedures shall also include the steps to be followed for breaking the specimens and examination of the fractured parts to determine the weld quality. The procedures shall also describe the corrective action to be taken in the event that the specimen test demonstrates weld quality that is not acceptable, including disposition of component welding accomplished by the welder since the last acceptable weldbreak test.

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3.22.3.2 Steel welding. The surfaces of steel parts to be welded shall abe free from rust, scale, paint, grease, and other foreign matter. Welds shall transmit stress without permanent deformation or failure when the part connected by the welds are subjected to proof and service loading. Parent materials, weld filler metals, and fabrication techniques shall be as required to enable the bridge to conform to the examination and test requirements specified in section 4. Parts to be joined by fillet welds shall be brought into as close contact as possible and in no event shall be separated by more than 1/8 inch unless appropriate bridging techniques are used.

3.22.3.2.1 Maraging steel welding. Maraging steel assemblies shall be welded using the GTAW or TIG process.

3.22.3.3 Care of welding electrodes. Care of welding electrodes specified for welding high-strength steel shall be as follows:

a. When a container of welding electrodes is opened, the entire contents shall be placed immediately in a baking oven, with electrodes placed not more than three layers deep on a tray, and baked either (1) for a minimum of 1 hour, but not more than 1-1/4 hours, at 800 °F; or (2) for a minimum of 4 hours at 400 °F.
b. The baked electrodes, while still warm, shall then be placed in a ventilated holding oven set at 250 to 300 °F.
c. Not more than a 1/2-hour supply of welding electrodes shall be taken form the holding oven at one time.
d. If electrodes are out of the holding oven in excess of 1/2 hour but less than 1 hour, they shall be returned to the holding oven for a period of not less than 1 hour.
e. If electrodes are out of the holding oven in excess of 1 hour, they shall be returned to the baking oven and rebaked as described under a.
f. Electrodes removed for use shall not be allowed to come in contact with any moist or wet surfaces. Electrodes that have been in contact with water, grease, or oil shall not be used.

3.22.4 Welders. Before assigning any welder to fabricate bridges covered by this specification, the contractor shall obtain certification that the welder has passed qualification tests as prescribed by the ASME Boiler and Pressure Vessel Code, Section IX, for the type of welding operations to be performed and that such qualification is effective as defined by the code. Test specimens shall be made on the same alloy, gage and temper of material with the same filler materials and in the same position as will be used in production.

3.22.5 Machine work. Tolerances and gages for metal fits shall conform to the limits specified on the applicable end product drawings.

3.22.6 Jigs and fixtures. Fabrication, assembly and installation of all bridge components (except minor parts) shall be accomplished with production tooling, jigs or fixtures, and parts shall be joined while held in position. Jigs or fixtures shall be designed to minimize distortion of the parts being welded and to assure that the configuration, alignment and location of parts meet the specified requirements. Subsequent straightening and thermal stress

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relieving is permitted to relieve distortions, provided the straightening process does not damage the component. Jigs and fixtures used to join maraging steel parts must be designed to account for the shrinkage of maraging steel during heating processes.

3.22.7 Machining of maraging steel. Specical care must be taken to insure that parts are not work hardend during machining, i.e., sharp cutting tools and correct speed feed rates.

3.22.8 Riveted connections. Components shall be bolted in position before riveting. Suitable jigs and fixtures shall be used to maintain alignment and tolerance. Rivets shall fill the holes in the joined members. Caulking or redriving of rivet heads shall not be permitted. Riveting adjacent to welding shall be done after welding. Rivet tension shall not be created by driving incorrectly fitted members. Rivet assemblies shall be of uniform quality and free from cracks, gaps, sharp edges, burrs, loose parts, or other defects which might render the assemblies unsuitable for its intended purpose. All rivets and components having the following defects shall be removed and replaced.

(1) Loose rivets.
(2) Clinched (bent) rivets.
(3) Rivets with a complete offset driven head.
(4) Deformed or open seams caused by pressure on rivet.
(5) Bulging skin caused by expanded rivets or trapped chips.
(6) Cut or marred skin around rivet heads.

3.22.9 Rivets. The rivet type, size, and material shall be as specified on the engineering drawing, parts list, or specification.

3.22.10 Rivet holes. Rivet hole diameters shall be within the limits specified on the drawings. Oversize, oblong, and irregular shaped holes shall be cause for rejection. All holes shall be drilled normal (at 90 degrees) to the working surface. All chips, burrs, and foreign material shall be removed from the mating surfaces before parts are riveted together.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an

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acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Disassembly inspection. Failure of any examination or test by the first article shall be cause for disassembly, in the presence of a Government representative, of the first article to the extent necessary to determine the cause of failure. Each disassembled part shall be examined in detail for compliance with this specification and referenced drawings in regard to materials, dimensions, tolerances, and workmanship. Parts not complying with such requirements shall be rejected and shall be cause for rejection of the first article. Reassembly with replacement parts and retesting shall be the responsibility of the contractor.

4.1.3 Parts and components. Parts and components detailed on the drawings shall be inspected in accordance with the quality assurance provisions (QAP) shown on the drawings. The drawings specify the characteristics requiring QAP inspection, the sampling plan, and the basis for acceptance and rejection (see 6.6 and 6.7.1).

4.2 Classification of inspections. Inspections shall be classified as follows:

a. First article (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection of packaging (see 4.7).

4.3 First article inspection.

4.3.1 Examination. The first article bridge or separately procured components shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests.

4.3.2.1 Hydraulic system components. Prior to installation of the hydraulic system components into the first article bridge, the components shall be tested as specified in 4.5.2.1 through 4.5.2.7. Failure of any test shall be cause for performing the disassembly inspection specified in 4.1.2.

4.3.2.2 Assembled bridge. After successful completion of the tests specified in 4.3.2.1, the hydraulic system components shall be installed in the first article bridge. The assembled bridge shall be tested as specified in 4.5.2.8 thru 4.5.2.12. Failure of any test shall be cause for performing the disassembly inspection specified in 4.1.2.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination and tests for separately procured bridge components shall be in accordance with MIL-STD-105, table I and table IIa. A lot shall be accepted when 0 defects are found and rejected when 1 or more defects are found.

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4.4.2 Examination.

4.4.2.1 Individual. Each bridge shall be examined for the defects specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.4.2.2 Samples. Samples of separately procured bridge components selected in accordance with 4.4.1 shall be examined for the applicable defects specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.4.3 Tests.

4.4.3.1 Individual. Each bridge shall be tested as specified in 4.5.2.1 through 4.5.2.12. Failure of any test shall be cause for rejection.

4.4.3.2 Samples. Samples of separately procured components selected in accordance with 4.4.1 shall be subjected to the applicable test specified in 4.5.2.1 through 4.5.2.9. Failure of any test shall be cause for rejection.

4.5 Inspection schedule.

4.5.1 Examination. The bridge or separately procured components as specified shall be examined as specified herein for the following defects as applicable:

Major

101. Any part or component not in accordance with QAP requirements as shown on the end product drawing.
102. Any dimension other than those identified on the QAP requirements shown on the drawings not as specified.
103. Members not true to dimension and line or containing twists, bends, or other imperfections.
104. Weld defects such as base metal cracks adjacent to or behind welds, cracks in weld metal, undercutting of base metal, lack of fusion with parent metal or between parts, porosity, lack of penetration, lack of prescribed fit, incorrect preparation, slag inclusion, spatter, or incorrect size.
105. Qualification of welders or weld procedure not as specified.
106. Welding not performed in accordance with qualified procedure.
107. Welding electrodes or care of welding electrodes not as specified.
108. Material not as specified.
109. Materials are not corrosion resistant or treated to be made corrosion resistant for the applicable storage and operating environment.
110. Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other.
111. Contractor does not have documentation available for identification of material, or material finishes, or treatment.
112. Used, rebuilt, or remanufactured components, pieces or parts incorporated in the bridge.
113. Hydraulic oil not as specified.
114. Hydraulic system cleanliness not as specified.
115. Threads not as specified.
116. Commercial components not as specified.
117. Components not interchangeable with respect to installation, performance, and quality.

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118. Safety not as specified.
119. Human factors not as specified.
120. Specification-control items or qualification of specification-control items not as specified.
121. Fittings and hydraulic hose not as specified.
122. Leaks in hydraulic lines.
123. Quick-disconnect couplings not as specified.
124. Age of hose not as specified.
125. Components missing or not as specified.
126. Workmanship not as specified.
127. Metal fabrication not as specified.
128. Machine work not as specified.

Minor

201. Cleaning, treatment and painting not as specified.
202. Lubrication and lubrication fittings not as specified.
203. Identification markings and hose assembly identification missing or incorrect.
204. Jigs and fixtures not as specified.
205. Stenciling not as specified.

4.5.2 Tests.

4.5.2.1 Test conditions. All test shall be conducted at an ambient air temperature of from +25 to +100 °F, with the hydraulic oil temperature between +25 and +150 °F. Hydraulic oil conforming to MIL-H-6083, shall be used for all tests. Prior to assembly of hydraulic components for test or installation on bridge all hydraulic lines and hydraulic components shall be free of any foreign matter, such as metal chips and filings that may in any way contaminate the oil or damage the system. All equipment shall be operated in accordance with the applicable technical manual.

4.5.2.2 Test schedule. All tests shall be conducted in accordance with table II and in the specified sequence.

Table II. Test schedule.

First article	Quality conformance	Test	Test paragraph	Requirement paragraph
1	2	3	4	5
X	X	Test conditions.	4.5.2.1	
X	X	Limbering up of hydraulic cylinder.	4.5.2.3	
X	X	Hydraulic cylinder pressure.	4.5.2.4	3.5.1
X	X	Bridge articulation system pressure.	4.5.2.5	3.5.2, 3.5.3,
X	X	Relief valves and counterbalance valves.	4.5.2.6	3.5.4
X	X	Internal leakage.	4.5.2.7	3.5.5
X	X	Post-test hydraulic cleanliness	4.5.2.8	3.7

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Table II. Test schedule, Cont.

First article	Quality conformance	Test	Test paragraph	Requirement paragraph
1	2	3	4	5
X	X	Operational performance.	4.5.2.9	3.8
X	-	Reliability and maintainability.	4.5.2.10	3.14, 3.15
X	-	Slinging and tiedown provisions.	4.5.2.11	3.16
X	-	Rail impact test.	4.5.2.12	3.20

4.5.2.3 Limbering of hydraulic cylinder. Prior to subjecting a cylinder to the tests specified herein, the cylinder shall be cycled through 25 complete cycles. One cycle shall consist of extension and retraction of the rod through full travel without bottoming. The cycling shall be done by either hydraulic pressure or mechanical means.

4.5.2.4 Hydraulic cylinder pressure. Place the cylinder in the test frame (see 3.13) so that the piston is restrained form bearing against either cylinder head. In accordance with drawing 13226E5306, notes 13 and 14, subject the cylinder to 4500 psi, +0 psi -200 psi, for 5 minutes. Nonconformance to 3.5.1 shall constitute failure of this test.

4.5.2.5 Bridge articulation system pressure. Subject the hydraulic hose and hydraulic cylinder assembly to a pressure between 4450 psi and 4550 psi for at least 1 minute. Evidence of external leakage, permanent deformation or nonconformance to 3.5.2 or 3.5.3 shall constitute failure of this test.

4.5.2.6 Relief valves and counterbalance valves. Each valve shall be pressurized until the valve opens. Valves shall operate at the pressure specified on the applicable end product drawing (± 50 psi). Opening of the valve at any pressure below this range or failure to open before this range is exceeded shall constitute failure of this test. After passing the above test, the pressure shall be reduced to 100 psi below valve operating pressure. Failure of the valve to close (as indicated by an internal leakage of no more that 50 drops a minute) before this pressure is reached or nonconformance to 3.5.4 shall constitute failure of this test.

4.5.2.7 Internal leakage. During the hydraulic cylinder pressure test, leakage shall be measured, as specified on drawing 13226E5306 note 13. Leakage in excess of the rate specified shall constitute failure of this test.

4.5.2.8 Post-test hydraulic cleanliness. After completion of the operational performance test and prior to delivery of the bridge to the Government, a sample of hydraulic fluid shall be taken from one of the quick-disconnect couplings and tested for the particle count specified in table I. If the particle count exceeds the values in table I, the system shall be flushed until contamination is reduced to the acceptable level. Inability to conform to the cleanliness levels specified in table I shall constitute failure of this test.

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4.5.2.9 Operational performance. The bridge shall be launched using the pendant control. Following launching, the bridge shall be retrieved from the far end. Failure to conform to 3.8 shall constitute failure of this test.

4.5.2.10 Reliability/maintainability. Using the mean-time-between-failure (MTBF) specified in 3.14, the bridge (see 3.13) shall be tested as specified herein. Using the maintenance ratio of man-hours of maintenance to hours of operation performed for the bridge as specified in 3.15, the maintainability shall be determined as specified herein. Maintenance shall be performed in accordance with the technical manual, and at the end of each day's mission, any damage to the bridge shall be repaired prior to the next day's mission. Failure to meet the requirements of 3.8 shall be cause for rejection. Failure to meet all performance requirements of this specification, when such failure is due to faulty material or workmanship, shall be cause for rejection and shall be cause for refusal by the Government to continue acceptance of production units until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on all units previously produced. Any such material or workmanship deficiencies will be considered prima facie evidence that all units accepted prior to the completion of this test are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.5.2.10.1 Reliability-maintainability test cycle. Reliability-maintainability testing shall be conducted in cycles, each cycle shall consist of the following:

a. 26 Km of transporting.
b. launch.
c. 10 class 3 vehicle crossings.
d. retrieve.

4.5.2.10.2 Reliability. Test the bridge for 86 cycles in accordance with Test Plan XVII of MIL-STD-781. For the purpose of reliability computations, a mission failure is defined as follows:

a. Any malfunction which would cause inability to complete a bridging cycle as specified in 4.5.2.10.1.
b. Any impending malfunction which would require maintenance that cannot be deferred until the end of the mission and would create a safety hazard to personnel or equipment as defined by MIL-STD-882.
C. Any critical structural member failure.

Simultaneous related malfunctions are considered one failure. Malfunctions which do not affect mission performance are not considered mission failures.

4.5.2.10.3 Maintainability. The maintenance manhours required to perform all corrective preventive maintenance occuring during the reliability test (see 4.5.2.10.2) shall be recorded and classified as to whether it was scheduled or unscheduled. The MR requirement(s) of 3.15 shall be assessed using the following formula:

$$MR = \frac{\text{Total scheduled maint. mhrs + total unscheduled maint. mhrs}}{\text{Total operating time}}$$

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Operator/crew checks and services are excluded from this ratio. Failure to meet the requirements as specified in 3.15 shall constitute failure of this test.

4.5.2.11 Slinging and tiedown provisions. With the bridge anchored by means other that the tiedown or slinging provision being tested, apply the static loads specified in MIL-STD-209 to each provision. Weld failure, permanent deformation of the bay or its tiedown or slinging provisions, or breakage of the provisions shall constitute failure of this test.

4.5.2.12 Rail impact test. With the bridge mounted on the launcher, the system shall undergo a rail impact test as specified in MIL-STD-810. Weld failure, breakage, or permanent deformation of the bridge shall constitute failure of this test.

4.6 Inspection comparison. The Government may select bridges at any time during the contract production period and subject these bridges to the examination specified in 4.5.1 and the tests specified in 4.5.2 to determine conformance to the requirements of this specification. The inspection will be performed by the Government, at a site selected by the Government, on bridges selected at random from those which have been accepted by the Government and will not include the previously inspected first article bridge. In addition to any test specified as part of the inspection comparison, the Government reserves the right to conduct any and all other tests contained in this specification as part of the inspection comparison, and failure of such additional tests shall have the same effect as failure of those tests specified in inspection comparison.

4.6.1 Inspection failure. Failure of an inspection comparison bridge to meet any requirement specified herein during and as a result of the examination and tests specified in 4.6 shall be cause for rejection of the inspection comparison bridge bay and shall be cause for refusal by the Government to continue acceptance of production bridges until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on bridges previously accepted and produced under the contract. Any deficiencies found as a result of the inspection comparison will be considered prima facie evidence that all bridges accepted prior to the completion of inspection comparison are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.7 Inspection of packaging. Each bridge shall be examined for the following defects: Any defects shall be corrected to comply with the requirements of section 5 before shipment of the bridge.

129. Unfinished surfaces not preserved as specified.
130. Machined surfaces not preserved as specified.
131. Hydraulic system no filled with specified fluid.
132. Hydraulic hoses not secured to bridge.
133. Bridge no packed as specified.
134. Marking incorrect, illegible or missing.

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5. PACKAGING

5.1 Preservation and packing (see 6.8).

5.1.1. Unfinished surfaces. Before any surfaces are coated with any preservative(s) they shall be cleaned and dried in accordance with MIL-P-116. Unpainted or unfinished exterior metal surfaces requiring the application of a contact preservative in accordance with MIL-P-116 shall be coated with type P-1 preservative.

5.1.2 Machined surfaces. Before any surfaces are coated with any preservative(s), they shall be cleaned and dried in accordance with MIL-P-116. Exposed machined metal surfaces for which a grease coating has not been specified on the drawings shall be coated type P-2 or type P-11 preservative.

5.1.3 Hydraulic system. The hydraulic system shall be filled to its operating level with the specified hydraulic fluid. Connections on hydraulic hoses shall be fitted with caps or covered with waterproof barrier material, secured with tape conforming with PPP-T-60, type III or IV to prevent entrance of dirt and water. Hydraulic hoses shall be secured to the launcher with plastic ties or similar material to prevent flailing about during shipment.

5.2 Packing (see 6.8). The bridge shall be folded and secured in the transport position. At least two steel straps, class I type I, (heavy duty), in accordance with QQ-S-781 shall encircle the bridge to prevent accidental opening during shipment and storage. Edge protectors shall be used at the points of contact between strap and bridge corners.

5.3 Marking (see 6.8). Marking for shipment and storage shall be in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The bridge is intended to be used with a launcher conforming to drawing 13226E5250. The bridge is capable of supporting class 30 vehicles over gaps up to 23 meters (75 feet).

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of the specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
c. When first article bridge or separately procured components are required and the number of each type of bridge component required (see 3.2).
d. When Government-loaned property is not required (see 3.13).

6.3 First article. When a first article inspection is required, the item(s) should be a preproduction model. The first article should consist of one or more units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of the

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first article test results and disposition of the first articles. Invitation for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Provisioning. The contracting officer should include provisioning requirements for repair parts and maintenance tools as necessary (including any special tools), and instructions on shipment of bridges. A suggested paragraph is as follows:

"Shipment of bridges shall include repair parts, maintenance tools, operational instructions, and accessories, unless exceptions are provided elsewhere in the contract."

6.5 Government-loaned property. The contracting officer should arrange to loan the property specified in 3.13.

6.6 Quality assurance provisions (QAP). The contracting office should require the contractors to maintain records of all QAP inspections. A suggested paragraph is as follows:

"The contractor shall maintain complete records of all examinations and tests performed to verify the requirements of classified QAP characteristics. The records shall include, as a minimum, lot size, sample size,drawing requirements, actual measurement, number and type of deficiencies found, quantity approved, quantity rejected, and corrective action taken when applicable."

6.7 Definitions.

6.7.1 Quality assurance provisions (QAP). A QAP is a contractual requirement that supplements section 4 of this specification. QAP indicates minimum requirements which must be inspected on the end product drawings to verify the design objectives of the product and assure interchangeability of repair parts.

6.8 Preservation and packing. Levels of preservation and packing have not been included as the preservation specified is the minimum acceptable for any level of protection, and no packing is required. For purposes of preservation/packing level marking, the bridges shall be marked A/A.

6.9 Subject term (key word) listing.

Bridge system
Double-fold bridge
Scissoring type bridge

Custodian:
Army - ME

Preparing activity:
Army - ME

Project 5420-A349

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-B-53107(ME)	900223

3. DOCUMENT TITLE
Bridge, Light Assault: 23 Meter Span MLC 30

4. NATURE OF CHANGE (*Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.*)

5. REASON FOR RECOMMENDATION

[redacted]

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (*Include Area Code*) (1) Commercial	(2) AUTOVON
	(703) 664-5717	354-5717
c. ADDRESS (*Include Zip Code*) US Army Belvoir RDE Center ATTN: STRBE-TSE Ft. Belvoir, VA 22060-5606	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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